Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 50%. We keep your highest score.

Next item →

1. Which techniques can be considered alternatives to hyperparameter optimization for improving model performance? 1 / 1 point

- [x] Data augmentation

  Correct
  Correct! Data augmentation can improve model robustness and performance.

- [x] Feature engineering

  Correct
  Correct! Feature engineering can enhance model performance by providing better input data.

- [ ] Ignoring model validation
- [x] Transfer learning

  Correct
  Correct! Transfer learning is a viable alternative for improving model performance without extensive hyperparameter tuning.

- [ ] Using deeper networks without tuning

2. Which of the following best describes the role of hyperparameters in a machine learning model? 1 / 1 point

- ○ Hyperparameters are the weights of the connections between neurons in a neural network.
- ○ Hyperparameters are the output predictions made by the model after training.
- ◉ Hyperparameters are used to tune the performance of a model based on the training data.
- ○ Hyperparameters are the data inputs to the model during training.

  Correct
  Correct! Hyperparameters are tuned to optimize the model's performance.

3. Select the correct statements regarding the architecture of VGG models. 1 / 1 point

- [x] They employ fully connected layers at the end of the network.

  Correct
  Good job! VGG models end with fully connected layers.

- [ ] They are based on recurrent neural networks.
- [ ] They integrate batch normalization at every layer.
- [x] VGG models use very small (3x3) convolutional filters.

  Correct
  Correct! VGG models are notable for their use of small 3x3 convolutional filters.

- [x] They use a varying number of filters across different layers.

  Correct
  Correct! VGG models increase the number of filters as you go deeper into the network.

4. Which process involves adjusting the weights of a pretrained model based on new data? 1 / 1 point

- ◉ Fine-tuning
- ○ Prediction
- ○ Feature selection
- ○ Data augmentation

  Correct
  Correct! Fine-tuning adjusts the weights of a pretrained model to better fit the new data.

5. Which scenarios are appropriate for using pretrained deep learning models? 1 / 1 point

- [x] Feature extraction for a new task

  Correct
  Well done! Pretrained models are excellent for feature extraction in new tasks.

- [ ] Training a model from scratch with a massive dataset
- [ ] Replacing traditional machine learning algorithms
- [x] Predicting image classes with limited data

  Correct
  Correct! Pretrained models can perform well even with limited data.

- [x] Transfer learning for a related task

  Correct
  Correct! Transfer learning is a common use case for pretrained models.